Immune Checkpoint Inhibitor-Associated Glomerular Disease: A Systematic Review and Descriptive Analysis

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Supplementary Material

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PRISMA CHECKLIST

Section/topic	#	Checklist item	Reported on page #/section
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	Title
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	Abstract
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	Introduction
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	Introduction
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	N/A
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	Methods, Study Selection
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	Methods, Literature Sources
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Supplementary Materials, Appendix A

Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	Methods, Study Selection
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	Methods, Data Abstraction
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	Methods, Data Abstraction
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	Methods, Causality Assessment and Assessment of Quality and Bias
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	Methods, Statistical Analyses
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	Methods, Statistical Analyses

Section/topic	#	Checklist item	Reported on page # /section
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	Methods, Causality Assessment and Assessment of Quality and Bias
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	N/A
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	Results, Search Results
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Results, Aggregate Data of Patients with ICI-Associated Glomerular Disease
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Results, Search Results
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	N/A
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Results, Tables 1-4
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	Supplementary Materials, Appendix C
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	N/A
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	Discussion
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	Discussion (Limitations)

Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	Conclusions
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	N/A

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

For more information, visit: **www.prisma-statement.org**.

Supplementary Material – Literature Search Strategy

Date: February 11, 2020

Information Specialist:
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Research Question

[What literature is available on the immune-related adverse effects of immune checkpoint inhibitors on the glomerulus?]

Description

The search strategy takes the format of Immune Checkpoint Inhibitors AND Glomerular Disease.

Limits

The search has been limited to English language articles, published from 2010 to the present. Animal-only studies have also been removed.

Number of results per database

Databases Searched	**Date of Search**	**Number of Results**
All Ovid Medline <1946 - present>	February 11, 2020	148
EBM Reviews - Cochrane Central Register of Controlled Trials <January 2020>	February 11, 2020	36
Embase Classic+Embase <1947 to 2020 February 10>	February 11, 2020	690
	Total	874
	Total (After deduplication)	**740**

Search Strategies

Medline

1 Immunotherapy/ae [Adverse Effects] (1844)
2 Ipilimumab/ (1730)
3 Nivolumab/ (2053)
4 (check?point adj2 inhibit*).tw,kf. (9753)
5 (check?point adj2 block*).tw,kf. (3674)
6 immuno?therap*.tw,kf. (85004)
7 CTLA-4 blockade*.tw,kf. (419)
8 Ipilimumab.tw,kf. (2891)
9 Yervoy.tw,kf. (50)
10 PD-1 inhibitor*.tw,kf. (889)
11 Anti-PD-1.tw,kf. (3093)
12 Pembrolizumab.tw,kf. (3036)
13 Keytruda.tw,kf. (65)
14 PD-L1 inhibitor*.tw,kf. (653)
15 Anti-PD-L1.tw,kf. (1199)
16 Atezolizumab.tw,kf. (765)
17 Tecentriq.tw,kf. (22)
18 Avelumab.tw,kf. (332)
19 Bavencio.tw,kf. (11)
20 Durvalumab.tw,kf. (361)
21 Imfinzi.tw,kf. (5)
22 Nivolumab.tw,kf. (4043)
23 Opdivo.tw,kf. (61)
24 Cemiplimab.tw,kf. (41)
25 Libtayo.tw,kf. (4)
26 1 or 2 or 3 or 4 or 5 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 (21197)
27 exp Acute Kidney Injury/ (45075)
28 exp Nephritis/ (73226)
29 exp Amyloidosis/ (24097)

30 exp Nephrosis/ (20332)
31 (acute kidney adj5 (injur* or insufficienc* or failure*)).tw,kf. (22516)
32 AKI.tw,kf. (11754)
33 (acute renal adj5 (injur* or insufficienc* or failure*)).tw,kf. (25905)
34 nephritis.tw,kf. (24524)
35 glomerul*.tw,kf. (126844)
36 amyloidos?s.tw,kf. (23328)
37 nephropath*.tw,kf. (57071)
38 minimal change disease*.tw,kf. (1593)
39 acute tubular necrosis.tw,kf. (3312)
40 nephrotoxicit*.tw,kf. (17554)
41 kidney toxicit*.tw,kf. (605)
42 nephros?s.tw,kf. (4443)
43 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 or 35 or 36 or 37 or 38 or 39 or 40 or 41 or 42 (312615)
44 26 and 43 (184)
45 limit 44 to english language (176)
46 exp Animals/ not Humans/ (4672574)
47 45 not 46 (167)
48 limit 47 to yr="2010 -Current" (148)

Cochrane Central

1 Immunotherapy/ae [Adverse Effects] (0)
2 Ipilimumab/ (138)
3 Nivolumab/ (362)
4 (check?point adj2 inhibit*).tw. (1000)
5 (check?point adj2 block*).tw. (233)
6 immuno?therap*.tw. (8505)
7 CTLA-4 blockade*.tw. (32)
8 Ipilimumab.tw. (1052)
9 Yervoy.tw. (45)
10 PD-1 inhibitor*.tw. (180)
11 Anti-PD-1.tw. (621)
12 Pembrolizumab.tw. (1260)
13 Keytruda.tw. (49)
14 PD-L1 inhibitor*.tw. (96)
15 Anti-PD-L1.tw. (363)

16 Atezolizumab.tw. (587)
17 Tecentriq.tw. (41)
18 Avelumab.tw. (168)
19 Bavencio.tw. (3)
20 Durvalumab.tw. (404)
21 Imfinzi.tw. (1)
22 Nivolumab.tw. (1504)
23 Opdivo.tw. (67)
24 Cemiplimab.tw. (28)
25 Libtayo.tw. (0)
26 1 or 2 or 3 or 4 or 5 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 (4570)
27 exp Acute Kidney Injury/ (1358)
28 exp Nephritis/ (1189)
29 exp Amyloidosis/ (153)
30 exp Nephrosis/ (380)
31 (acute kidney adj5 (injur* or insufficienc* or failure*)).tw. (2569)
32 AKI.tw. (1430)
33 (acute renal adj5 (injur* or insufficienc* or failure*)).tw. (1529)
34 nephritis.tw. (1083)
35 glomerul*.tw. (8758)
36 amyloidos?s.tw. (452)
37 nephropath*.tw. (6498)
38 minimal change disease*.tw. (60)
39 acute tubular necrosis.tw. (122)
40 nephrotoxicit*.tw. (1581)
41 kidney toxicit*.tw. (39)
42 nephros?s.tw. (63)
43 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 or 35 or 36 or 37 or 38 or 39 or 40 or 41 or 42 (20841)
44 26 and 43 (42)
45 limit 44 to english language (37)
46 exp Animals/ not Humans/ (10)
47 45 not 46 (37)
48 limit 47 to yr="2010 -Current" (36)

Embase

1 immunotherapy/ae [Adverse Drug Reaction] (153)
2 ipilimumab/ (12776)
3 pembrolizumab/ (12889)
4 atezolizumab/ (4239)
5 avelumab/ (1874)
6 durvalumab/ (3083)
7 nivolumab/ (15078)
8 cemiplimab/ (188)
9 (check?point adj2 inhibit*).tw,kw. (18610)
10 (check?point adj2 block*).tw,kw. (6985)
11 CTLA-4 blockade*.tw,kw. (761)
12 Ipilimumab.tw,kw. (6494)
13 Yervoy.tw,kw. (759)
14 PD-1 inhibitor*.tw,kw. (2053)
15 Anti-PD-1.tw,kw. (7844)
16 Pembrolizumab.tw,kw. (7534)
17 Keytruda.tw,kw. (687)
18 PD-L1 inhibitor*.tw,kw. (1353)
19 Anti-PD-L1.tw,kw. (3224)
20 Atezolizumab.tw,kw. (1975)
21 Tecentriq.tw,kw. (224)
22 Avelumab.tw,kw. (780)
23 Bavencio.tw,kw. (108)
24 Durvalumab.tw,kw. (1191)
25 Imfinzi.tw,kw. (112)
26 Nivolumab.tw,kw. (9438)
27 Opdivo.tw,kw. (620)
28 Cemiplimab.tw,kw. (99)
29 Libtayo.tw,kw. (23)
30 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26 or 27 or 28 or 29 (48835)
31 acute kidney failure/ (85419)
32 acute kidney tubule necrosis/ (5093)
33 exp Nephritis/ (124747)
34 AA Amyloidosis/ (372)
35 exp Nephrosis/ (64320)
36 Immunoglobulin A nephropathy/ (11053)

37 exp Glomerulopathy/ (78263)

38 (acute kidney adj5 (injur* or insufficienc* or failure*)).tw,kw. (37022)

39 AKI.tw,kw. (20746)

40 (acute renal adj5 (injur* or insufficienc* or failure*)).tw,kw. (38058)

41 nephritis.tw,kw. (36870)

42 glomerul*.tw,kw. (180187)

43 amyloidos?s.tw,kw. (34197)

44 nephropath*.tw,kw. (82202)

45 minimal change disease*.tw,kw. (2299)

46 acute tubular necrosis.tw,kw. (4969)

47 nephrotoxicit*.tw,kw. (24462)

48 kidney toxicit*.tw,kw. (882)

49 nephros?s.tw,kw. (6250)

50 31 or 32 or 33 or 34 or 35 or 36 or 37 or 38 or 39 or 40 or 41 or 42 or 43 or 44 or 45 or 46 or 47 or 48 or 49 (493470)

51 30 and 50 (980)

52 limit 51 to english language (970)

53 animals/ not humans/ (1371006)

54 52 not 53 (970)

55 limit 54 to yr="2010 -Current" (945)

56 limit 55 to embase (690)

Table S1 – Summary of All Reported Cases of ICI-Associated Glomerular Disease and Causality Assessment

Reference	GN Category	Age	Sex	Cancer Category	Immune Checkpoint Inhibitor(s) Received	GN Category	WHO-UMC Causality Category
Ashour [33]	C3 GN	68	M	Melanoma	Pembrolizumab	C3 GN	Certain
Bickel[34]	MCD	62	M	Other	Pembrolizumab	MCD	Possible
Cho[35]	Pauci-immune	NA	NA	Other	Pembrolizumab	Pauci-immune	Probable
Cortazaar[36]	Pauci-immune	41	M	Lung	Nivolumab	Pauci-immune	Unclassifiable
Cortazaar[36]	MCD	NA	NA	NA	NA	MCD	Unclassifiable
Cortazaar[36]	Anti-GBM	57	M	Melanoma	Ipilumumab + Nivolumab	Anti-GBM	Unclassifiable
Cortazaar[36]	C3 GN	NA	NA	NA	NA	C3 GN	Unclassifiable
Cortazar[19]	TMA	58	M	Melanoma	Ipilimumab	TMA	Probable
Cruz-Whitley[67]	C3 GN	75	F	Lymphoma	Nivolumab	C3 GN	Probable
Daanen[53]	FSGS	62	M	RCC	Nivolumab	FSGS	Probable
Fadel[57]	Lupus-like Nephritis	64	M	Melanoma	Ipilimumab	Lupus-like Nephritis	Probable
Gallan[37]	Pauci-immune	68	M	Melanoma	Nivolumab	Pauci-immune	Probable
Gallan[37]	Pauci-immune	71	F	Lung	Pembrolizumab	Pauci-immune	Probable
Gallan[37]	Pauci-immune	75	F	Lung	Nivolumab	Pauci-immune	Probable
Gallan[37]	Pauci-immune	63	F	Melanoma	nivolumab	Pauci-immune	Probable
Gao[38]	MCD	40	M	Lymphoma	SHR-1210 (anti-PD-1)	MCD	Possible
Heo[39]	Pauci-immune	56	M	Other	Pembrolizumab	Pauci-immune	Probable
Irifuku[40]	Immune-complex GN	59	M	RCC	Nivolumab	Immune-complex GN	Probable
Izzedine[41]	MCD	NA	NA	Melanoma	Pembrolizumab	MCD	Probable
Izzedine[41]	MCD	NA	NA	Other	Pembrolizumab	MCD	Probable
Jung[42]	Immune-complex GN	70	M	RCC	Nivolumab	Immune-complex GN	Probable
Kidd[43]	MCD	55	M	Melanoma	Ipilimumab	MCD	Probable
Kishi[44]	IgA	72	M	Lung	Nivolumab	IgA	Probable
Kitchlu[45]	MCD	43	M	Lymphoma	pembrolizumab	MCD	Probable
Kitchlu[45]	MCD	45	M	Melanoma	Ipilimumab	MCD	Certain
Lapman[54]	AA Amyloidosis	42	M	Other	Pembrolizumab	AA Amyloidosis	Probable
Lapman[54]	AA Amyloidosis	75	M	Lung	Nivolumab	AA Amyloidosis	Probable
Lapman[54]	AA Amyloidosis	81	M	Melanoma	Nivolumab	Amyloidosis	Probable
Lemoine[46]	Pauci-immune	70	M	Melanoma	Ipilimumab	Pauci-immune	Probable
Mamlouk[47]	Pauci-immune	41	M	Lung	Nivolumab	Pauci-immune	Probable
Mamlouk[47]	Pauci-immune	75	M	RCC	Tremelimumab	Pauci-immune	Probable
Mamlouk[47]	Pauci-immune	69	F	Melanoma	Nivolumab and Ipilimumab	Pauci-immune	Probable
Mamlouk[47]	IgA	69	M	Melanoma	Nivolumab and Ipilimumab	IgA	Probable
Mamlouk[47]	IgA	50	F	Melanoma	Pembrolizumab	IgA	Probable

Mamlouk[47]	Membranous	60	F	RCC	Nivolumab	Membranous	Possible
Mamlouk[47]	C3 GN	61	F	Other	Pembrolizumab	C3 GN	Probable
Mamlouk[47]	FSGS	74	M	RCC	Nivolumab	FSGS	Probable
Mamlouk[47]	AA Amyloidosis	63	M	Other	Pembrolizumab	AA Amyloidosis	Probable
Oki[51]	IgA	75	F	Lung	Pembrolizumab	IgA	Probable
Person[48]	Pauci-immune	55	M	Melanoma	Nivolumab and Ipilimumab	Pauci-immune	Probable
Saito[52]	MCD	79	M	Lung	pembrolizumab	MCD	Possible
Sammartino[55]	Anti-GBM	50	M	Melanoma	Tremelimumab	Anti-GBM	Probable
Seethapathy[20]	TMA	60	F	NA	NA	TMA	Unclassifiable
Takahashi[49]	Anti-GBM	74	M	Lung	Nivolumab	Anti-GBM	Probable
Ville[50]	C3 GN	73	M	Lung	Pembrolizumab	C3 GN	Probable

GN: Glomerulonephritis; WHO-UMC: World Health Organization Uppsala Monitoring Center Case Causality Category

Table S2 – Quality Assessment of Case Reports and Series of Immune Checkpoint Inhibitor-Associated Glomerular Disease

Reference	Patients	Selection	Ascertainment		Causality				Reporting	
	(n)	Was exposure adequately ascertained?	Was exposure adequately ascertained?	Was the outcome adequately ascertained?	Were other alternative causes ruled out?	Was there a challenge/re-challenge phenomenon?	Was there a dose-response effect?	Was the follow up long enough for outcomes to occur?	Is the case described with sufficient details?*	Quality
Ashour, et al 2019	1	No	Yes	Yes	Yes	Yes	No	Yes	Yes	Good 6
Bickel, et al 2016	1	No	Yes	Yes	Yes	No	No	Yes	Yes	Fair 5
Cruz-Whitley, et al 2020	1	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Good 6
Daanen, et al 2017	1	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Good 6
Fadel, et al 2009	1	No	Yes	Yes	Yes	No	No	Yes	Yes	Fair 5
Gallan, et al 2019	4	No	Yes	Yes	Yes	No	No	Yes	Yes	Fair 5
Gao, et al 2018	1	No	Yes	Yes	Yes	No	No	Yes	Yes	Fair 5
Heo, et al 2017	1	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Good 6
Irifuku, et al 2020	1	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Good 7
Jung, et al	1	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Fair 5
Kidd, et al	1	No	No	Yes	Yes	No	No	Yes	Yes	Fair 5
Kishi, et al	1	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Fair 5
Kitchlu, et al 2017	2	No	Yes	Yes	Yes	Yes	No	Yes	Yes	Good 6
Lemoine, et al	1	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Good 6
Oki, et al	1	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Fair 5
Person, et al	1	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Good 7
Saito, et al 2019	1	No	Yes	Yes	Yes	Yes	No	Yes	Yes	Good 6
Takahashi, et al 2018	1	No	Yes	Yes	Yes	No	No	Yes	Yes	Fair 5
Ville, et al 2020	1	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Good 6
Sammartino, et al 2009	1	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Good 6
Lapman, et 2020	3	No	Yes	Yes	Yes	No	No	Yes	Yes	Fair 5

Izzedine, et al 2019	2	No	Yes	Yes	Yes	Yes	No	Yes	Yes	Good 6
Mamlouk, et al	9	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Good 6
Cortazar, et al 2016	1	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Good 6
Cho, et al	1	Yes	Yes	Yes	No	No	No	No	Yes	Poor 3

*****to allow other investigators to replicated research or allow practitioners to make inferences related to their own practice.

Table S3 – Newcastle-Ottawa Scale for Non-Randomized Studies for Immune Checkpoint Inhibitor-Associated Glomerular Disease

Reference	Selection	Comparability	Outcome	Score
Cortazar, et al. 2020	★★★★	★	★★★	8
Seethapathy, et al	★★★★	★★	★★★	9

Table S4 – Exploratory Univariate Logistic Regression for Predictor of Endstage Renal Disease or Death in Patients with Glomerular Disease Associated with Immune Checkpoint Inhibitors

Predictor	Odds Ratio (95% CI)	P value
Glomerular Lesion		
Podocytopathy	Ref	
Pauci-immune GN/Renal Vasculitis	0.58 (0.10, 3.51)	0.77
Other Lesion	1.00 (0.22, 4.81)	
ICI Class		
PD-1/PD-L1	Ref	
CTLA-4	3.01 (0.43, 25.4)	0.48
Combination	0.67 (0.03, 5.29)	
Peak proteinuria (per 1 g/d increment)	0.96 (0.83, 1.09)	0.54
Peak serum creatinine (per 1 mg/dL increment)	1.06 (0.78, 1.46)	0.70
Corticosteroid Use	1.12 (0.10, 23.3)	0.93